

RampEMS for Stroke-Stable Brush Pressure Modulation in Calligraphy

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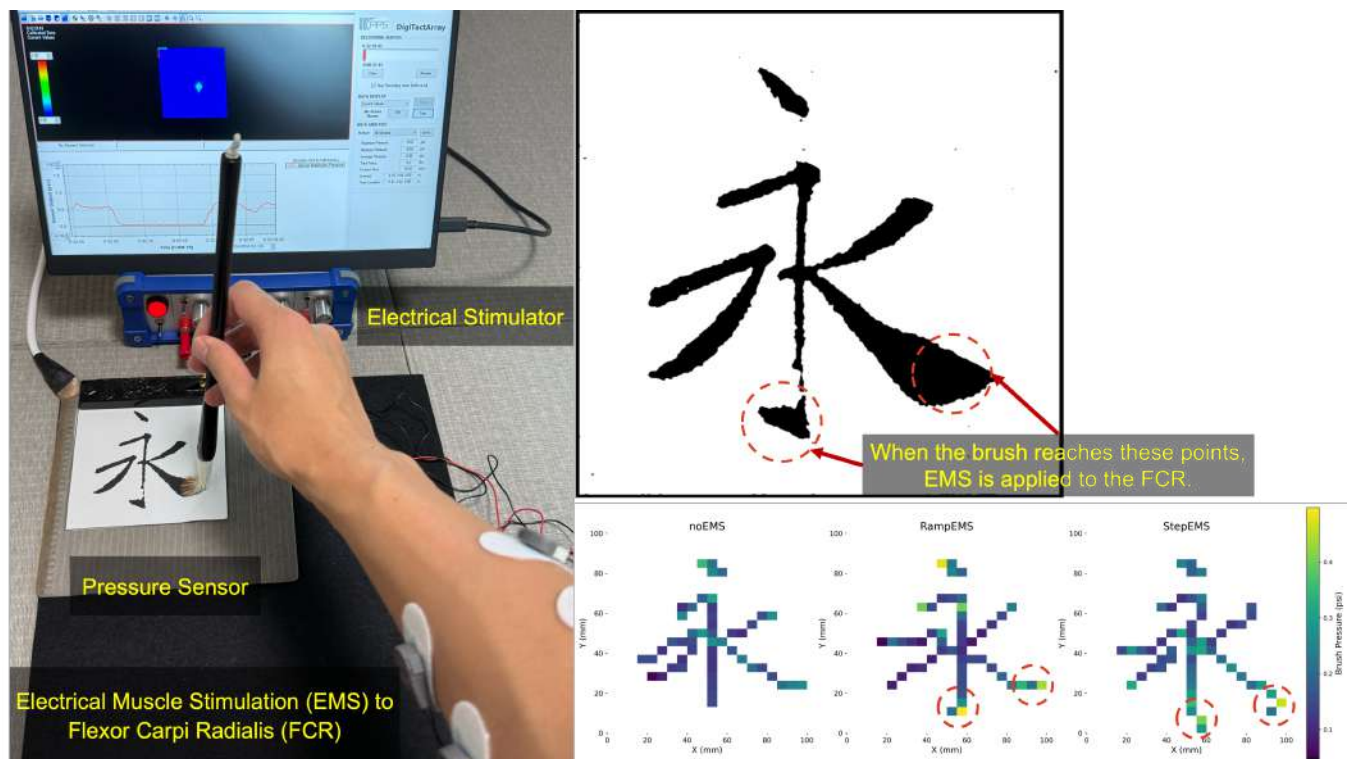


Figure 1: RampEMS-based calligraphy support system. A pressure sensor tracks brush position and pressure in real time and triggers electrical muscle stimulation (EMS) of the forearm muscles at predefined target points. RampEMS ramps pulse width to modulate brush pressure while preserving stroke trajectories, whereas StepEMS applies the maximum pulse width immediately, often distorting strokes near the stimulation points.

Abstract

Electrical muscle stimulation (EMS) has been explored in HCI to support motor activities such as sports training and musical performance. Because contraction strength increases with stimulation intensity, EMS can modulate applied force and may support fine motor tasks such as brush-pressure control in calligraphy. However, conventional step-type EMS (StepEMS) applies the target intensity



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abruptly, often eliciting an involuntary onset twitch that distorts stroke trajectories. We propose RampEMS, which ramps pulse width to the target level over 1 s. This onset profile suppresses sudden movements while maintaining brush-pressure modulation. In a user study, we compared RampEMS and StepEMS in a calligraphy task and measured brush-pressure change and stroke deviation. Results indicate that RampEMS enables pressure modulation while significantly reducing stroke disturbance relative to StepEMS.

CCS Concepts

• **Human-centered computing** → **Haptic devices**.

Keywords

Electrical Muscle Stimulation, Force Control, Ramp Stimulation

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1 Introduction

Electrical muscle stimulation (EMS) induces muscle contractions by delivering electrical pulses to the body. EMS has been explored in HCI as a lightweight wearable actuator for motor support in domains such as sports training [8, 17, 21], musical performance [4, 12, 18], and virtual reality [3, 14, 15]. Because contraction strength increases with stimulation intensity (e.g., pulse width), EMS can modulate users' applied force and provide graded force feedback [11, 13, 16].

However, conventional step-type EMS (StepEMS) steps to the target intensity abruptly and often elicits an onset twitch—a sudden involuntary movement immediately after stimulation begins [1, 7]. While such a twitch can be useful for quickly moving a limb, it disrupts tasks that require a stable trajectory. Lowering intensity mitigates twitches but also reduces torque, creating a trade-off between strong force modulation and motion stability.

We propose *RampEMS*, which ramps stimulation intensity by increasing pulse width to the target over one second rather than applying it instantly. This gradual onset suppresses twitches while preserving effective force modulation. Although ramp-up profiles are common in rehabilitation to improve comfort [6, 20, 23], we highlight their value for handwriting tasks that require stable fine-motor control.

We demonstrate RampEMS in calligraphy, where brush-pressure control is essential for stroke thickness and expressiveness [9, 19, 22]. Our system (Figure 1) senses brush position and pressure, and triggers EMS delivery to forearm muscles to increase or decrease brush pressure during writing. Step-type EMS often distorts strokes due to reflexive wrist motion at stimulation onset, whereas RampEMS enables substantial pressure modulation while preserving stroke trajectories. We evaluate RampEMS in a user study and show its broader applicability to tasks that require changing force without altering movement.

2 RampEMS

Ramped stimulation profiles are commonly used in rehabilitation to smooth contraction onset and reduce discomfort, typically with 1–3 s rise times [6, 20, 23]. We adapt this idea to brush-pressure modulation in calligraphy.

2.1 Algorithm

RampEMS increases pulse width from 0 to the maximum pulse width ($W_{\max}$) over a rise time, whereas StepEMS applies $W_{\max}$ immediately. Because contraction strength increases with pulse width, StepEMS can strongly modulate force but often elicits an onset twitch that distorts strokes. RampEMS reduces such artifacts by allowing users to adapt to a gradual increase in intensity.

To explore ramp functions for controlling pulse width in RampEMS, we conducted a formative study, approved by our institutional ethics board, with 14 participants using a pen tablet (Wacom CTL-672). EMS targeted the flexor carpi radialis (FCR) and the extensor digitorum (ED) to increase and decrease pen pressure, respectively. We tested linear, logarithmic, exponential, and sigmoid ramps, and selected a logarithmic profile for this work based on the formative study outcomes. Guided by prior work (1–3 s) and our pilot observations, we set the rise time to 1 s.

Finally, the pulse width trajectory of RampEMS, $W(t)$, is defined as follows:

$$W(t) = \begin{cases} W_{\max} \frac{\ln\left(1 + k \frac{t - t_{\text{start}}}{t_{\text{end}} - t_{\text{start}}}\right)}{\ln(1 + k)} & (t_{\text{start}} \leq t \leq t_{\text{end}}), \\ W_{\max} & (t > t_{\text{end}}). \end{cases} \quad (1)$$

where t denotes time (s), t_{start} denotes stimulation onset, t_{end} denotes the time at which the maximum pulse width is reached, $W_{\max}$ is the maximum pulse width, and k controls the curvature of the logarithmic rise. In this study, we set $t_{\text{end}} - t_{\text{start}} = 1$ s, $W_{\max} = 200 \mu\text{s}$, and $k = 9$.

2.2 Implementation

We implemented a calligraphy support system incorporating RampEMS. The system combined a pressure-sensor array for tracking brush position and pressure in real time with a programmable electrical stimulator with real-time pulse-width control. Brush data were obtained via the sensor API, and EMS was triggered when the brush entered predefined stimulation regions. For RampEMS, the pulse width increased gradually from 0 to $W_{\max}$ over 1 s.

For pressure sensing, we used a flexible pressure sensor made of conductive fabric (Pressure Profile Systems, Inc., Conformable TactArray). The full-scale range was 3.02 psi, with a resolution of 0.005 psi and a scan rate of 6.7 Hz. The array consisted of 1024 sensors (32×32), each 5×5 mm, enabling simultaneous measurement of pressure distribution over a 160×160 mm area.

EMS was delivered using a compact custom 4-channel stimulator with an integrated microcontroller (Arduino) for real-time parameter control. The stimulator provided a constant voltage of 0–60 V. EMS parameters were set to 200 Hz frequency and 0–200 μs pulse width. Disposable electrodes (Nihon Kohden, F-150M) were placed on the FCR and ED: stimulation of FCR was intended to increase

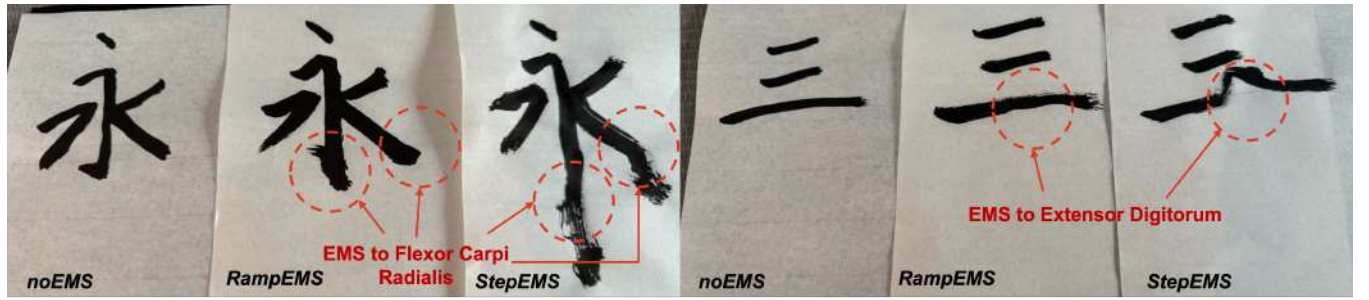


Figure 2: Representative examples of the characters Ei and San written under three conditions (noEMS, RampEMS, StepEMS). For Ei, EMS targeted the flexor carpi radialis (FCR) at two predefined points, whereas for San, EMS targeted the extensor digitorum (ED) at one predefined point. RampEMS reduced stroke distortion relative to StepEMS at the stimulation points.

brush pressure, while stimulation of ED was intended to decrease it.

Figure 2 shows representative calligraphic outputs when EMS is delivered to the FCR (Ei) or the ED (San). Compared with noEMS, RampEMS modulates stroke thickness with less visible trajectory distortion than StepEMS. By contrast, StepEMS induces abrupt wrist motion at stimulation onset, resulting in larger stroke distortion.

3 User Study

We evaluated whether RampEMS enables brush-pressure modulation while reducing stroke distortion, compared with StepEMS. Participants traced two printed characters (Ei, San) under three conditions (noEMS, RampEMS, StepEMS). We recorded brush pressure and stroke trajectories, and collected usability and workload ratings and open-ended feedback. The study was approved by our institutional ethics board.

3.1 Participants

Twenty adults participated (14 female, 6 male; ages 20s–50s; $M = 42.7$, $SD = 15.7$). Fourteen participants reported no prior calligraphy experience, and fourteen reported no prior experience with EMS. Eighteen participants were right-handed and two were left-handed; all participants performed the task with their right hand for consistency. All participants provided written informed consent.

3.2 Apparatus and Calibration

Brush position and pressure were captured using a pressure-sensor array, and EMS was delivered using a programmable stimulator with real-time pulse-width control. Disposable electrodes were placed on the flexor carpi radialis (FCR) and the extensor digitorum (ED). We calibrated stimulation voltage for each participant using non-painful abrupt-onset stimulation by gradually increasing voltage until clear wrist flexion/extension was observed (typically 40–50 V). Frequency and maximum pulse width were fixed at 200 Hz and 200 μ s, respectively.

3.3 Procedure

3.3.1 Characters for Writing. We selected Ei for the pressure-increase task and San for the pressure-decrease task (Figure 2). This choice

was informed by a preliminary measurement in which a calligraphy expert wrote seven candidate characters while we recorded brush pressure with the sensor. Ei contained segments with clear pressure increases, whereas San contained segments with clear pressure decreases, making them suitable for our manipulation. For Ei, the pressure increase was most pronounced in the hook and right-falling strokes; for San, the pressure decrease was most evident during the third horizontal stroke.

3.3.2 Task. Participants held the brush using a traditional calligraphy grip: the shaft was supported by the thumb, index, and middle fingers and stabilized from below by the ring and little fingers. They maintained the brush in an upright orientation. To reduce variability in writing style, participants traced reference characters printed on paper (110 $\times$ 110 mm) rather than writing freely. They were instructed to write slowly while maintaining light, approximately constant brush pressure.

3.3.3 Conditions. Each participant performed the writing task under three conditions: noEMS, RampEMS, and StepEMS. For each character, three trials were conducted per condition (nine trials per character; 18 trials total). The order of conditions was counterbalanced across participants. Before the measured trials, participants practiced each condition for approximately one minute. After completing the nine trials for Ei, participants took a 10-minute break before proceeding to the trials for San.

3.3.4 EMS Method. For Ei, EMS targeted the FCR during two stroke segments (the hook and right-falling strokes; Figure 2). Brush position was tracked in real time using the pressure-sensor array, and stimulation was triggered when the brush entered predefined target points/regions. In the RampEMS condition, pulse width ramped from 0 to 200 μ s over 1 s and was then held at 200 μ s until the end of the stroke (i.e., brush lift). In the StepEMS condition, stimulation stepped to 200 μ s immediately at onset and was maintained until brush lift. To standardize stimulation duration, participants were instructed to keep the brush on the paper for at least two seconds after stimulation onset before lifting it, based on preliminary measurements with expert calligraphers.

For San, EMS targeted the ED during the middle third (between one-third and two-thirds) of the third horizontal stroke. The stimulation pattern was the same as for Ei, and the duration corresponded

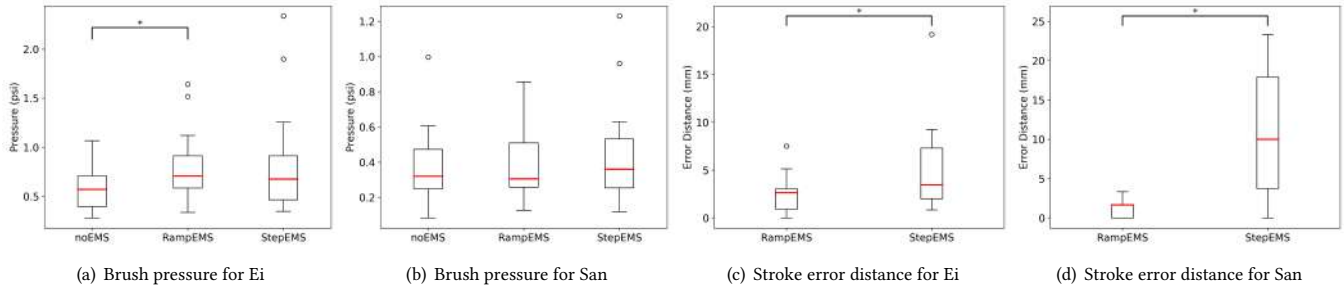


Figure 3: Boxplots of brush pressure and stroke error distance. When writing Ei, brush pressure under RampEMS was significantly higher than under noEMS. For both characters, stroke error distance was significantly lower under RampEMS than under StepEMS.

to the time required for the brush to traverse this segment (approximately 1–2 s). Stimulation stopped once the brush exited the stimulation region.

3.3.5 Objective Measures. During each trial, we recorded brush pressure and brush coordinates. For brush pressure, we computed a representative value within the target segments: the maximum pressure for Ei and the minimum pressure for San. We then averaged the representative value across the three trials per condition. For Ei, which included two target segments, we computed segment-wise values and averaged them.

We compared brush-pressure values across the three conditions (noEMS, RampEMS, StepEMS) to test whether EMS modulated brush pressure and whether the onset profile affected the achieved pressure. To quantify stroke distortion, we computed stroke error distance as the deviation of traced coordinates under RampEMS and StepEMS relative to the corresponding noEMS trajectories within the target segments. We compared stroke error distances between RampEMS and StepEMS to assess whether the stimulation profile affected stroke distortion.

3.3.6 Subjective Measures. After the experiment, we assessed usability and workload for RampEMS and StepEMS using the System Usability Scale (SUS) [2] and a subset of NASA-TLX items [10]. SUS was scored following established guidelines and converted to a 0–100 scale. For NASA-TLX, participants rated four subscales (Mental Demand, Physical Demand, Effort, and Frustration) on a 7-point Likert scale. Participants also provided open-ended comments comparing the two stimulation methods.

3.4 Analysis

We analyzed brush pressure, stroke error distance, SUS, and NASA-TLX ratings. We first assessed normality using the Shapiro–Wilk test. When normality was satisfied, we used paired t -tests; otherwise, we used Wilcoxon signed-rank tests. Effect sizes were reported as Cohen’s d for parametric tests and r for non-parametric tests [5]. To control for multiple comparisons, we applied Holm’s method.

3.5 Results

Two participants showed unusually high baseline brush pressure in the noEMS condition ($> \text{mean} + 2 \text{ SD}$). To avoid undue influence,

we excluded these participants from all objective analyses of brush pressure and stroke error distance ($N = 18$). Subjective ratings (SUS and NASA-TLX) include all 20 participants.

3.5.1 Brush Pressure. For Ei, brush pressure was higher under RampEMS than under noEMS (Figure 3a; mean $\pm$ SD: noEMS 0.57 ± 0.21 psi, RampEMS 0.79 ± 0.35 psi, StepEMS 0.82 ± 0.55 psi). A Wilcoxon signed-rank test showed a significant increase from noEMS to RampEMS ($W = 28.5$, $p = 0.036$, $r = 0.67$). No significant differences were found for noEMS vs. StepEMS ($W = 44$, $p = 0.15$, $r = 0.49$) or RampEMS vs. StepEMS ($W = 69$, $p = 0.50$, $r = 0.19$).

For San, representative pressure (minimum within the target segment) was similar across conditions (Figure 3b; noEMS 0.37 ± 0.21 psi, RampEMS 0.38 ± 0.19 psi, StepEMS 0.43 ± 0.28 psi), with no significant pairwise differences (noEMS vs. RampEMS: $W = 84$, $p = 0.97$, $r = 0.02$; noEMS vs. StepEMS: $W = 60$, $p = 0.85$, $r = 0.30$; RampEMS vs. StepEMS: $W = 66$, $p = 0.85$, $r = 0.23$). Although stimulation targeted the ED to reduce brush pressure, we did not observe a decrease at the group level, potentially due to a floor effect (baseline pressure under noEMS was already low). As an exploratory analysis, the five participants with the highest noEMS pressure all showed decreased pressure under RampEMS (noEMS 0.64 ± 0.21 psi vs. RampEMS 0.47 ± 0.24 psi).

3.5.2 Stroke Error Distance. Stroke error distance (deviation from the noEMS trajectory) was lower with RampEMS than with StepEMS for both characters (Figure 3c,d). For Ei, RampEMS yielded 2.49 ± 1.82 mm vs. StepEMS 5.08 ± 4.49 mm ($W = 17$, $p = 0.0083$, $r = 0.75$). For San, RampEMS yielded 1.39 ± 1.31 mm vs. StepEMS 11.1 ± 8.10 mm ($W = 0$, $p = 0.00065$, $r = 1.00$). These results indicate that a ramped onset reduced stroke distortion relative to abrupt onset, regardless of whether stimulation targeted the FCR (Ei) or the ED (San).

3.5.3 SUS and NASA-TLX. SUS was significantly higher for RampEMS than for StepEMS (mean $\pm$ SD: RampEMS 69 ± 13 vs. StepEMS 52 ± 18 ; $t(19) = 5.45$, $p = 2.91 \times 10^{-5}$, $d = 1.2$), indicating higher perceived usability for RampEMS in the calligraphy task. NASA-TLX ratings were lower for RampEMS than for StepEMS across all four subscales: Mental Demand (2.6 ± 1.6 vs. 3.3 ± 1.7 ; $W = 31.5$, $p = 0.038$, $r = 0.59$), Physical Demand (2.9 ± 1.5 vs.

4.7 ± 1.7 ; $W = 8$, $p = 0.0024$, $r = 0.90$), Effort (2.7 ± 1.6 vs. 3.9 ± 1.6 ; $W = 27.5$, $p = 0.038$, $r = 0.64$), and Frustration (2.1 ± 1.3 vs. 3.5 ± 1.5 ; $W = 0$, $p = 0.0014$, $r = 1.00$). Overall, RampEMS reduced perceived workload compared with StepEMS.

3.5.4 Subjective Impressions. In post-experiment comments, many participants reported negative impressions of StepEMS. They described that it “disrupted the handwriting” (P02), was “difficult to write with and caused discomfort due to sudden stimulation” (P04), and created “a strong feeling of being controlled” (P13). Several participants also noted that the abrupt onset induced hand twitches that made it difficult to draw straight lines (P19), and caused surprise or psychological burden due to its suddenness (P11, P15).

By contrast, participants often described RampEMS as easier to write with, less uncomfortable, and more natural. For example, they noted that RampEMS “allowed gradual muscle response, making writing easier and less uncomfortable” (P08), and that it “felt smoother and gave a sense of control and reassurance” (P15, P16, P18).

4 Conclusion

We presented RampEMS, a ramped-onset EMS profile for stable brush-pressure modulation in calligraphy. By ramping pulse width to the target over 1 s, RampEMS suppresses involuntary onset twitches commonly observed with StepEMS while preserving force modulation. We implemented a calligraphy support system that triggers EMS based on real-time brush position and pressure. In a user study, RampEMS significantly reduced stroke distortion compared with StepEMS and improved subjective usability and workload ratings. Future work will extend RampEMS to other fine motor tasks that require changing force without altering movement.

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